Two Surveys, Two Biases

AP Statistics · Topic 3.4 (CED 1.12) · B: 2026-10-01 · E: 2026-10-09 · TEACHER KEY

CED TETHER

- **Skill 1.C** — Describe an appropriate method for gathering and representing data.
- **EU DAT-2** — The way we collect data influences what we can and cannot say about a population.
- **LO DAT-2.E** — Identify potential sources of bias in sampling methods. [Skill 1.C]
- **EK DAT-2.E.1** — Bias occurs when certain responses are systematically favored over others.
- **EK DAT-2.E.2** — When a sample is comprised entirely of volunteers or people who choose to participate, the sample will typically not be representative of the population (voluntary response bias).
- **EK DAT-2.E.3** — When part of the population has a reduced chance of being included in the sample, the sample will typically not be representative of the population (undercoverage bias).
- **EK DAT-2.E.4** — Individuals chosen for the sample for whom data cannot be obtained (or who refuse to respond) may differ from those for whom data can be obtained (nonresponse bias).
- **EK DAT-2.E.5** — Problems in the data gathering instrument or process result in response bias. Examples include questions that are confusing or leading (question wording bias) and self-reported responses.
- **EK DAT-2.E.6** — Non-random sampling methods (for example, samples chosen by convenience or voluntary response) introduce potential for bias because they do not use chance to select the individuals.

Essential question: How do response behavior and question wording limit the estimate a survey can support?

DOK-3 skill: 4.A · **FRQ pattern:** `name-the-bias-direction-and-fix`

DOK rationale: Part (c) adjudicates two competing claims about accuracy by separating response rate from response quality, uses evidence to reject an unsupported closeness ranking, and matches a distinct repair to each bias.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK: 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Read both council claims without confirming either.
- Begin the video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and require a separate diagnosis for each survey.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose one estimate or neither and cite one survey detail.
- Complete the video follow-along about undercoverage, nonresponse, voluntary response, and response bias.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Who is missing from the mail results even though the original sample was random?

- Can you know whether the nonresponders oppose the fee more or less?
- Which words in the phone question could move an answer, and in what direction?
- Does knowing one bias's likely direction tell you which final estimate is closer?

Adult role
Solo teacher. Circulate 5 pairs. Require students to connect each flaw to a mechanism and a repair; do not accept a percentage choice based only on which sample is larger.

[WARN] WATCH FOR

- Confusing nonresponse with voluntary response; the households were selected but did not reply.
- Assuming an 80 percent response rate protects against loaded wording.
- Claiming the exact size of either bias or ranking closeness without truth data.

SCORING PART (c) — E / P / I

Expected elements

- **rejects-unsupported-ranking** (required) — States that the information does not establish which percentage is closer to the true town opinion
- **uses-mail-evidence** (required) — Connects the 24 percent response rate to possible differences between responders and nonresponders
- **uses-phone-evidence** (required) — Connects the loaded wording to response bias and its likely upward effect on opposition
- **matches-two-fixes** (required) — Gives a nonresponse-focused follow-up for mail and neutral wording for phone without discarding random selection

E	Rejects both closeness claims, uses the 24 percent response rate and loaded wording to explain why, and gives a distinct workable fix for each survey.
P	Correctly distrusts both estimates and identifies both biases but omits the unsupported-ranking point, a direction explanation, or one matched fix.
I	Declares one estimate closer solely from sample size or response rate, says random selection eliminates all bias, or proposes no bias-specific repair.

Common mistakes

- Choosing 62% solely because 240 people answered
- Saying the mail survey is unbiased because the original 500 were randomly chosen
- Assuming mail nonresponse must increase opposition even though its direction is unknown
- Fixing the phone survey by calling more numbers but keeping the loaded wording

[STAR] TODAY'S PROBLEM — annotated

Suppose a town studies support for a park fee.

Mail survey: A neutral question is mailed to 500 randomly chosen households. Only 120 return it; 78% of respondents oppose the fee.

Phone survey: The town calls 300 random phone numbers. Of 240 people who answer, 62% oppose after hearing, *Do you support a NEW FEE that would tax park users?*

One council member says 62% must be closer to the truth because more people answered. Another says 78% must be closer because the households were randomly chosen.

Park-fee surveys

	Selected	Replied	Oppose the fee
Mail	500	120	78%
Phone	300	240	62%

FIRST TAKE (5 min)

Which estimate, if either, would you report to the town? Name one detail behind your choice.
Key: Not graded. Either number may feel safer; the goal is to separate known flaws from unknowable bias size.

Rules callout "BIAS FAVORS SOME RESPONSES (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Compute the response rate for each survey. Name the main source of bias in each.

Key: Mail response rate: $120/500 = 24\%$; the main problem is nonresponse bias. Phone response rate: $240/300 = 80\%$; the main problem is response bias from leading or loaded wording.

DOK 2 **(b)** Explain how each bias could change the reported percent opposing the fee. State a likely direction when the information supports one.

Key: Mail nonrespondents may hold systematically different opinions from respondents, so 78% could be too high or too low; the direction is not known from the data. Calling the fee NEW and saying it would tax users likely pushes people toward opposition, so 62% is likely too high as an estimate of opposition to neutrally described policy.

DOK 3 **(c)** Decide whether either council member has established which estimate is closer to the town's true opinion. Cite the 24% response rate and the loaded wording in your reasoning, explain why neither estimate is trustworthy as is, and propose one design-specific fix for each survey.

Key: Neither member has established which estimate is closer because the true town proportion and the sizes of the two biases are unknown. Randomly choosing the mail sample does not rescue a 24% response rate if responders differ from nonresponders. An 80% phone response rate does not rescue a loaded question that can systematically increase opposition. Fix the mail survey with repeated neutral follow-ups or mixed contact modes aimed at selected nonresponders; fix the phone survey by asking a neutral, balanced question while keeping the random-number selection.

EXIT

One thing the video changed about my first take: ______________________